

Activity on PostGresql

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Course code: DSA0503

Subject : Query Processing for Data Science

Activity using SQLite Database:

1. Create the Department Table (Parent Table) with attributes Department_ID

INTEGER PRIMARY KEY, Department_Name TEXT

code

```
import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

cur = conn.cursor()

cur.execute("""
CREATE TABLE IF NOT EXISTS Department (
    Department_ID INTEGER PRIMARY KEY,
    Department_Name TEXT
)
""")

conn.commit()

print("Department table created successfully")

cur.close()
```

```
conn.close()
```

```
|===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1.py ===  
|Department table created successfully
```

2. Create t

he Student Table (Child Table) with the following attributes:

Student_ID INTEGER PRIMARY KEY,

Name TEXT,

Age INTEGER,

Department_ID INTEGER,

CGPA REAL,

FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)

Code:

```
import psycopg2
```

```
conn = psycopg2.connect(
```

```
    host="localhost",
```

```
    database="college_db",
```

```
    user="postgres",
```

```
    password="123456",
```

```
    port="5432"
```

```
)
```

```
cur = conn.cursor()
```

```
cur.execute("""
```

```
CREATE TABLE IF NOT EXISTS Student (
```

```
    Student_ID INTEGER PRIMARY KEY,
```

```
    Name TEXT,
```

```
    Age INTEGER,
```

```
    Department_ID INTEGER,
```

```
    CGPA REAL,
```

```
    FOREIGN KEY (Department_ID)
```

```
    REFERENCES Department(Department_ID)
```

```

)
"""
conn.commit()
print("Student table created successfully")
cur.close()
conn.close()

===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,2.py =
Student table created successfully

```

3.Display students from CSE department.

Code

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

cur = conn.cursor()
cur.execute("""
INSERT INTO Department VALUES
(1, 'CSE'),
(2, 'ECE'),
(3, 'MECH'),
(4, 'CIVIL')
ON CONFLICT (Department_ID) DO NOTHING
""")

cur.execute("""
INSERT INTO Student VALUES

```

```

(101, 'Arun', 20, 1, 8.5),
(102, 'Priya', 21, 1, 9.0),
(103, 'Rahul', 20, 2, 7.8),
(104, 'Meena', 22, 3, 8.2)
ON CONFLICT (Student_ID) DO NOTHING
"""
conn.commit()

print("Sample records inserted successfully")

cur.close()

conn.close()

===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,3.py
Sample records inserted successfully

```

4.Count students in each department.

Code

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

cur = conn.cursor()

cur.execute("""
SELECT S.*
FROM Student S
JOIN Department D
ON S.Department_ID = D.Department_ID
WHERE D.Department_Name = 'CSE'

```

```

"""
for row in cur.fetchall():
    print(row)
cur.close()
conn.close()

===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,4.py :
(101, 'Arun', 20, 1, 8.5)
(102, 'Priya', 21, 1, 9.0)

```

5. Update the foreign key. For eg., change the department of a student

Code

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

cur = conn.cursor()

cur.execute("""
SELECT D.Department_Name, COUNT(S.Student_ID)
FROM Department D
LEFT JOIN Student S
ON D.Department_ID = S.Department_ID
GROUP BY D.Department_ID, D.Department_Name
ORDER BY D.Department_ID
""")

for row in cur.fetchall():
    print(row)
cur.close()

```

```
conn.close()
```

```
===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,5.py
('CSE', 2)
('ECE', 1)
('MECH', 1)
('CIVIL', 0)
```

6. Delete a department

Code

```
import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

cur = conn.cursor()

ur.execute("""
UPDATE Student
SET Department_ID = 1
WHERE Student_ID = 103
""")

conn.commit()

print("Student department updated successfully")

cur.close()

conn.close()

===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,6.py
Student department updated successfully
```

7. Display each students name along with the corresponding department name using an SQL JOIN.

Code

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

cur = conn.cursor()
cur.execute("""
DELETE FROM Department
WHERE Department_ID = 4
""")

conn.commit()

print("Department deleted successfully")

cur.close()

conn.close()

| ===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,7.py :
| Department deleted successfully

```

8. Display the number of students in each department.

Code

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

```

```

cur = conn.cursor()
cur.execute("""
SELECT S.Name, D.Department_Name
FROM Student S
JOIN Department D
ON S.Department_ID = D.Department_ID
""")
for row in cur.fetchall():
    print(row)
cur.close()
conn.close()

===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,8.py
('Arun', 'CSE')
('Priya', 'CSE')
('Meena', 'MECH')
('Rahul', 'CSE')

```

9. Display the names of students belonging to the Computer Science department.

Code

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

cur = conn.cursor()
cur.execute("""
SELECT D.Department_Name, COUNT(S.Student_ID)
FROM Department D
LEFT JOIN Student S

```



```

ON D.Department_ID = S.Department_ID
GROUP BY D.Department_ID, D.Department_Name
ORDER BY D.Department_ID
"""
)
for row in cur.fetchall():
    print(row)
cur.close()
conn.close()

===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,9.py
('CSE', 3)
('ECE', 0)
('MECH', 1)

```

10. Update the department of a student by changing the foreign key value and display the updated record.

Code

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="123456",
    port="5432"
)

cur = conn.cursor()
cur.execute("""
SELECT S.Name
FROM Student S
JOIN Department D
ON S.Department_ID = D.Department_ID
WHERE D.Department_Name = 'CSE'

```

```

"""
for row in cur.fetchall():
    print(row[0])
cur.close()
conn.close()

===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,10.py
Arun
Priya
Rahul

===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,11.py =
(103, 'Rahul', 'CSE')

```

11. Display departments that currently have no students assigned using an appropriate SQL JOIN.

Code

```

import psycopg2

conn = psycopg2.connect(
    host="localhost",
    database="college_db",
    user="postgres",
    password="your_password",
    port="5432"
)

cur = conn.cursor()
cur.execute("""
SELECT D.Department_Name
FROM Department D
LEFT JOIN Student S
ON D.Department_ID = S.Department_ID
WHERE S.Student_ID IS NULL
""")

for row in cur.fetchall():

```

```
print(row[0])
```

```
cur.close()
```

```
conn.close()
```

```
===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/act 1,12.py ===  
ECE
```